This assignment will not be graded and is only for practice.

Key Points: 1

Two non-zero vectors u and v from an inner product space V are said to be orthogonal if $\langle u, v\rangle = 0$.
A set S of non-zero vectors is said to be an orthogonal set of an inner product space if elements of the set S are mutually orthogonal. i.e., $\langle v_i, v_j\rangle = 0$ for $i = j$, any $v_i, v_j \in S$.
Let $\{v_1, v_2, \ldots, v_n\}$ be an orthogonal set of vectors in an inner product space. Then $\{v_1, v_2, \ldots, v_n\}$ is a linearly independent set of vectors.
Orthogonal basis: Let V be an inner product space. A basis consisting of mutually orthogonal vectors is called an orthogonal basis. In other word, a maximal orthogonal set is a basis and is called an orthogonal basis.
An orthonormal set of vectors of an inner product space V is an orthogonal set of vectors such that the norm of each vector of the set is 1.
An orthonormal basis is an orthonormal set of vectors which form a basis.
Obtaining orthonormal set from orthogonal set: If $\gamma = \{v_1, v_2, \ldots, v_n\}$ is an orthogonal set of vectors then we can obtain an orthonormal set of vectors β from γ by dividing each vector by its norm i.e.,

$$\beta = \left\{\frac{v_1}{\|v_1\|}, \frac{v_2}{\|v_2\|}, \ldots, \frac{v_n}{\|v_n\|}\right\}.$$

Consider a set of vectors $S = \{(1, -1, -1), (1, 2, 0), (-1, 1, 1), (-1, \frac{1}{2}, -\frac{3}{2}), (0, 3, 1), (2, 1, 1), (0, 1, -1)\}$ from the inner product space $\mathbb{R}^3$ with usual inner product (i.e., the dot product).
Answer questions (1) and (2).

1) Which of the following subsets of the set S consist of mutually orthogonal vectors? ***1 point***

- [ ] $\{(1,-1,-1),(1,2,0),(-1,1,1)\}$
- [ ] $\{(1,-1,-1),(-1,\frac{1}{2},-\frac{3}{2})\}$
- [ ] $\{(1,-1,-1),(-1,\frac{1}{2},-\frac{3}{2}),(1,2,0)\}$
- [ ] S
- [ ] $\{(2,1,1),(1,-1,-1),(0,1,-1)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(1,-1,-1),(-1,\frac{1}{2},-\frac{3}{2})\}$
$\{(2,1,1),(1,-1,-1),(0,1,-1)\}$

2) Which of the following option(s) is (are) true? ***1 point***

- [ ] $\{(2,1,1),(0,1,-1)\}$ is an orthogonal basis of a vector space which is isomorphic to $\mathbb{R}^2$.
- [ ] $\{(2,1,1)\}$ is not an orthogonal basis of a vector space which is isomorphic to $\mathbb{R}$.
- [ ] $\{(1,-1,-1),(1,2,0)\}$ is an orthogonal basis of a vector space which is isomorphic to $\mathbb{R}^2$.
- [ ] $\{(-1,1,1),(-1,\frac{1}{2},-\frac{3}{2}),(0,3,1),\}$ is an orthogonal basis of $\mathbb{R}^3$.
- [ ] $\{\frac{(2,1,1)}{\sqrt{6}},\frac{(1,-1,-1)}{\sqrt{3}},\frac{(0,1,-1)}{\sqrt{2}}\}$ is an orthonormal basis of $\mathbb{R}^3$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(2,1,1),(0,1,-1)\}$ is an orthogonal basis of a vector space which is isomorphic to $\mathbb{R}^2$.

$\{\frac{(2,1,1)}{\sqrt{6}}, \frac{(1,-1,-1)}{\sqrt{3}}, \frac{(0,1,-1)}{\sqrt{2}}\}$ is an orthonormal basis of $\mathbb{R}^3$.

Key Points: 2

The projection of a vector to a subspace: Let V be an inner product space, $v \in V$ and $W \subseteq V$ be a subspace. Then the projection of v onto W is the vector in W, denoted by $proj_W(v)$, computed as follows:

Find an orthonormal basis $\{v_1, v_2, \ldots, v_n\}$ for W.
Then $proj_W(v) = \sum_{i=1}^{n} \langle v, v_i \rangle v_i$.

Remark: $proj_W(v)$ is independent of the chosen orthonormal basis.
Let V be an inner product space and W be a subspace. Then the projection of vectors in V to W is a linear transformation from V to V with image W.

3) Consider the vector space $\mathbb{R}^3$ with usual inner product. Which of the following vectors represents projection of the vector $u = (2, 1, -2)$ onto the subspace whose an orthonormal basis is $\{(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}), (\frac{-2}{3}, \frac{1}{3}, \frac{2}{3})\}$? ***1 point***

- $(\frac{14}{9}, \frac{-7}{9}, \frac{-14}{9})$
- (1,1,1)
- (2,1,-2)
- $(\sqrt{2}, 0, \sqrt{2})$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$(\frac{14}{9}, \frac{-7}{9}, \frac{-14}{9})$

4) Consider the vector space $\mathbb{R}^3$ with usual inner product. If (a, b, c) is the projection vector obtained from projecting the vector $(1, 0, 1)$ onto the subspace $\{(x, y, z) \mid x - y + z = 0, x, y, z \in \mathbb{R}\}$, then find the value $3(a + b + c)$.

Hint: Find any orthonormal basis of the subspace $\{(x, y, z) \mid x - y + z = 0, x, y, z \in \mathbb{R}\}$ and use the definition of the projection of a vector onto a subspace.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 4

1 point

Key Points: 3

Gram-schmidt process: The Gram-schmidt process is used to get an orthogonal basis from any given basis of a vector space. A slight modification yields an orthonormal basis.
Let $\gamma = \{v_1, v_2, \ldots, v_n\}$ be a given ordered basis of a vector space. The steps in the process are described below and yield an orthonormal basis $\beta = \{u_1, u_2, \ldots, u_n\}$.

Step-1 $w_1 = v_1$, and $u_1 = \frac{w_1}{\|w_1\|}$.

Step-2 $w_2 = v_2 - \langle v_2, u_1 \rangle u_1$ and $u_2 = \frac{w_2}{\|w_2\|}$.

Step-3 $w_3 = v_3 - \langle v_3, u_1 \rangle u_1 - \langle v_3, u_2 \rangle u_2$ and $u_3 = \frac{w_3}{\|w_3\|}$.

$\vdots$

Step-i $w_i = v_i - \sum_{j=1}^{i-1} \langle v_i, u_j \rangle u_j$ and $u_i = \frac{w_i}{\|w_i\|}$.

$\vdots$

Step-n $w_n = v_n - \sum_{j=1}^{n-1} \langle v_n, u_j \rangle u_j$ and $u_n = \frac{w_n}{\|w_n\|}$.

Let β be an ordered basis of the inner product space R^3 with usual inner product, where $\beta = \{(1, 1, -1), (1, 1, 0), (1, 0, 0)\}$ and $\gamma = \{u_1, u_2, u_3\}$ be the corresponding orthonormal basis obtained using the Gram-schmidt process. Use this information to answer the questions (5), (6) and (7).

5) If $u_1 = (a_1, b_1, c_1)$, then find the value of $\sqrt{3}(a_1 + b_1 + c_1)$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

6) If $u_2 = (a_2, b_2, c_2)$, then find the value of $\sqrt{6}(a_2 + b_2 + c_2)$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 4

1 point

7) If $u_3 = (a_3, b_3, c_3)$, then find the value of $\frac{1}{\sqrt{2}}(a_3 + b_3 + c_3)$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

Key Points: 6

Orthogonal transformation: Let V be an inner product space and T be a linear transformation from V to V. T is said to be an orthogonal transformation if

$$\langle Tv, Tw\rangle = \langle v, w\rangle, \ \forall v, w \in V$$

Orthogonal matrix: A square matrix A is called an orthogonal matrix if $AA^T = A^TA = I$(identity matrix)

8) Consider the following linear transformations $T_1, T_2 : \mathbb{R}^2 \to \mathbb{R}^2$, and $T_3, T_4 : \mathbb{R}^3 \to \mathbb{R}^2$ defined as **1 point**

$$T_1(x, y) = \left(\frac{x+y}{\sqrt{2}}, \frac{x-y}{\sqrt{2}}\right),$$

$$T_2(x, y) = (x + 3y, 2x - 2y),$$

$$T_3(x, y, z) = \left(\frac{x+z}{\sqrt{2}}, \frac{x+y\sqrt{3}+z}{\sqrt{3}}\right),$$

$$T_4(x, y, z) = (x + y + z, x + y),$$

where $\mathbb{R}^3$ and $\mathbb{R}^2$ are the inner product spaces with respect to the dot product. Which of the linear transformation is an orthogonal transformation?

- ○ T_1
- ○ T_2
- ○ T_3
- ○ T_4

No, the answer is incorrect.
Score: 0

Accepted Answers:

T_1

9) Which of the following option(s) is (are) true? **1 point**

- ☐ $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ is an orthogonal matrix.
- ☐ $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$ is an orthogonal matrix.
- ☐ $\begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 2 \\ -1 & -2 & 1 \end{bmatrix}$ is an orthogonal matrix

- [ ] $\frac{1}{\sqrt{2}}\begin{bmatrix}1 & 0 & 1\\ 0 & \sqrt{2} & 0\\ 1 & 0 & -1\end{bmatrix}$ is an orthogonal matrix.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}$ is an orthogonal matrix.

$\frac{1}{\sqrt{2}}\begin{bmatrix}1 & -1\\ 1 & 1\end{bmatrix}$ is an orthogonal matrix.

$\frac{1}{\sqrt{2}}\begin{bmatrix}1 & 0 & 1\\ 0 & \sqrt{2} & 0\\ 1 & 0 & -1\end{bmatrix}$ is an orthogonal matrix.

Your score is: 0/9